

# DoangJoo "Alan" Synn

Ph.D. Student in Computer Science · Georgia Institute of Technology

Advised by Prof. HyunJoo Oh and Prof. Sehoon Ha

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## Research Interests

I study **motion** as a computational medium across HCI and computer graphics. My research builds systems for designing, generating, and fabricating motion. I aim to make motion programmable and tangible across animated characters, mechanical artifacts, robotic systems, and AI-assisted creative tools.

Before starting my Ph.D., I spent eight years building software systems across AI systems, games, cloud platforms, and social robots.

## Education

**Georgia Tech**, Ph.D. in Computer Science

2022–Present

- **School of Interactive Computing**; advised by Profs. Oh & Ha
- Research: computational design, mechanical artifacts, and motion synthesis (HCI/Graphics)

**Korea University**, M.Eng. in ECE

2020–2022

- Thesis: *On Efficient Data Delivery for Machine Learning Systems*

**Korea University**, B.Eng. in EE

2012–2020

## Experience

**Ph.D. Research Intern, Dolby Laboratories**

2024–2025

- Built a time-series, multimodal datastore supporting low-latency media analytics in the Experience Delivery Lab.

**MLOps Engineering Intern, LG Uplus Corp.**

2023

- Led **DeepCrunch**, a model-compression library supporting multi-backend AI inference for efficient edge deployment.

**Research Scientist, Protopia AI Inc.**

2021–2022

- Built **Stained-Glass**, a privacy-preserving deep-learning framework enabling inference without raw-data exposure.

**Cloud Architect · AWS Educate Cloud Ambassador**

2020–2022

- Led engineering for four startup services, translating ambiguous product requirements into production backend systems.

**Server Engineer, Flysher Inc.**

2019

- Shipped backend features for four social-game titles; one reached Facebook's global top 10.

**System Engineer, AKA Intelligence (Musio)**

2017–2019

- Designed embedded Linux boards and firmware for **Musio**, a deployed social robot.

## Selected Publications

- 1 **D. Synn**, Z. Guo, S. Ha, and H. Oh, “MotionSmith: A Sketch-Based Design System for Automata Making,” *CHI*, Apr. 2026.
- 2 J. Yu, P. Garg, **D. Synn**, and H. Oh, “Tangible-MakeCode: Bridging Physical Coding Blocks with a Web-Based Programming Interface for Collaborative and Extensible Learning,” *CHI*, Apr. 2025.
- 3 **D. Synn**, X. Piao, J. Park, and J. Kim, “Distributed Training of Deep Learning Models Using Micro Batch Streaming,” *KTSDE*, Mar. 2023.
- 4 **D. Synn** and J. Kim, “Num Worker Tuner: An Automated Spawn Parameter Tuner for Multi-Processing DataLoaders,” *ACK*, Oct. 2021.
- 5 B. Ahn, **D. Synn**, M. Derkani, E. Ebrahimi, and H. Esmailzadeh, “Protopia AI: Taking on the Missing Link in AI Privacy and Data Protection,” *NeurIPS Demo*, 2021.